Unit 3

Ordinary Point, Singular Point. $\longrightarrow$

$$y'' + P(x)y' + Q(x)y = 0$$

① Ordinary Point $\Rightarrow$ Value of x which is x_0 at which $P(x)$ & $Q(x)$ will be finite. $\lim_{x\to x_0} P(x) \neq \infty$ & $\lim_{x\to x_0} Q(x) \neq \infty$

② Singular Point $\Rightarrow$ $P(x)$ & $Q(x)$ will be infinite. $\lim_{x\to x_0} P(x) = \infty$ & $\lim_{x\to x_0} Q(x) = \infty$

(i) Regular Singular Point $\longrightarrow$ ~~Value of x at which~~

$$\boxed{\lim_{x\to x_0} (x-x_0)^2 P(x) \neq \infty} \ \& \ \boxed{\lim_{x\to x_0} (x-x_0)^2 Q(x) \neq \infty}$$

[Anyone true then it satisfies]

(ii) Irregular Singular Point $\longrightarrow$

$$\boxed{\lim_{x\to x_0} (x-x_0)^2 P(x) = \infty} \ \& \ \boxed{\lim_{x\to x_0} (x-x_0)^2 Q(x) = \infty}$$

① $2x(x-2)^2 \dfrac{d^2y}{dx^2} + 3x\dfrac{dy}{dx} + (x-2)y = 0$

$$\frac{d^2y}{dx^2} + \frac{3x}{2x(x-2)^2}\frac{dy}{dx} + \frac{\cancel{(x-2)}}{2x(x-2)^{\cancel{2}}}y = 0$$

$$\frac{d^2y}{dx^2} + \underbrace{\frac{3}{2(x-2)^2}}_{P(x)}\frac{dy}{dx} + \underbrace{\frac{1}{2x(x-2)}}_{Q(x)}y \longrightarrow 2(x-2)^2 = 0 \ \boxed{x_0 = 2} \ \& \ 2x(x-2) = 0 \ \boxed{x_0 = 2}$$

$$\underbrace{\lim_{x\to 2} \frac{3}{2(x-2)^2} = \infty \ \& \ \lim_{x\to 2} \frac{1}{2(x)(x-2)} = \infty}_{\text{Singular Point}}$$

$$\lim_{x\to 2} \frac{3}{2(x-2)^{2}} \cancel{(x-2)} \Rightarrow \frac{1}{0} = \infty \longrightarrow \text{Irregular Sing Pt}$$

Q $x^3(x-2)\frac{d^2y}{dx^2} + x^3\frac{dy}{dx} + 6y = 0$

$$\frac{d^2y}{dx^2} + \underbrace{\frac{1}{x-2}}_{P(x)}\frac{dy}{dx} + \underbrace{\frac{6}{x^3(x-2)}}_{Q(x)}y = 0$$

$P(x)=0$

$\rightarrow \frac{1}{x-2} \Rightarrow \boxed{x_0=2}$ & $x^3(x-2)=0$ & $\boxed{x=0}$ $\boxed{x_0=2}$

$\lim_{x\to 2}\frac{1}{x-2} = \infty$ & $\lim_{x\to 2}\frac{6}{x^3(x-2)} = \infty$

Singular Pt $(x=2) \Rightarrow$

$\rightarrow \lim_{x\to 2}\frac{1}{\cancel{x-2}}\cancel{(x-2)} = \boxed{1} = \neq \infty$

$\lim_{x\to 2}\frac{6}{x^3\cancel{(x-2)}}\cancel{(x-2)} = \frac{6}{8} \Rightarrow \boxed{\frac{3}{4}} \neq \infty$

$\therefore$ Regular Sing Pt

Sing Pt $(x=0) \Rightarrow$

$\rightarrow \lim_{x\to 0}\frac{1}{(x-2)}(x) \rightarrow 0$

$\lim_{x\to 0}\frac{6}{x^3(x-2)}x \Rightarrow \boxed{\infty}$

$\therefore$ Irregular Sg Pt

Q $x=0 \rightarrow$ O.P & $x=1$ Regular Sing Pt $\longrightarrow$

$$(x^2-1)\frac{\partial^2 y}{\partial x^2} + x\frac{\partial y}{\partial x} - y = 0$$

$\rightarrow \frac{\partial^2 y}{\partial x^2} + \underbrace{\frac{x}{x^2-1}}_{P(x)}\frac{\partial y}{\partial x} - \underbrace{\frac{1}{x^2-1}}_{Q(x)}y = 0$ } $P(x)=0$

$\frac{1}{x^2-1} = 0 \rightarrow \boxed{x_0 = \pm 1}$

$Q(x) = \boxed{x_0 = \pm 1}$

$\lim_{x\to 0}\frac{x}{x^2-1} = 0$ & $\lim_{x\to 0}\left(\frac{-1}{x^2-1}\right) = 1$ } $\neq \infty$ = O.P

$\lim_{x\to 1}\frac{x}{x^2-1} = \infty$ (S.P) & $\Rightarrow \lim_{x\to 1}\frac{x}{x^2-1}(x-1) \rightarrow \lim_{x\to 1}\frac{x}{x+1} \Rightarrow \frac{1}{2} \neq \infty \Rightarrow$ Regular S.P

Dirac Delta Method →

$$L\{\delta(t)\} = 1$$

$$L\{\delta(t-a)\} = e^{-as}$$

$$L\{f(t)\,\delta(t-a)\} = e^{-as} f(a)$$

Q L.T of $\delta(t-4) \Rightarrow$

e^{-4s}

Q L.T of $t^3 \delta(t-4)$

$e^{-4s} \, 4^3$

$\boxed{64 e^{-4s}}$

$\because f(t) = t^3$

$f(a) = a^3$